

# Management Report

## Scalable Music Genre Classification and Audio Analytics Using the Free Music Archive Dataset

*Decision-Oriented Summary for Project Review*

Student: Julian Devonish  
Programme: MSc Applied Data Science  
Course: COMP6830 - Data Science Capstone II  
Supervisor: Dr. Sean Miller  
Prepared: June 18, 2026

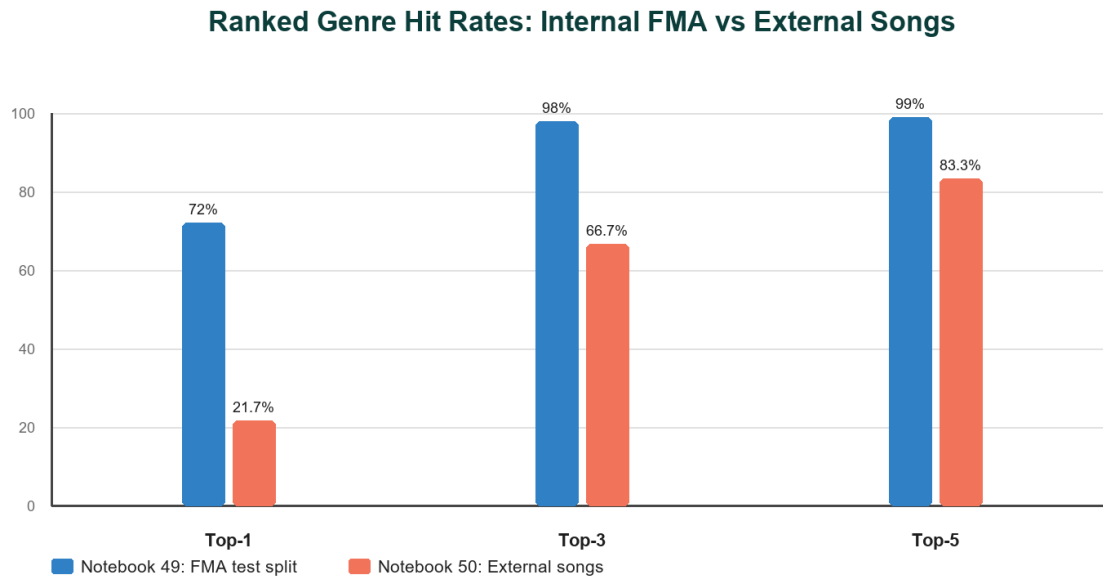
### Executive Summary

This management report summarises a six-month capstone project that built a ranked music genre prediction pipeline using the Free Music Archive dataset. The work combined structured metadata, engineered features, raw audio windows, machine learning models, deep learning models, and hybrid fusion. The most useful management conclusion is direct: the system is ready to demonstrate as a ranked genre suggestion tool, not as an unsupervised production tagger.

The reason for that position comes from the final evidence. Notebook 49 produced strong internal FMA test results, including Top-1 hit-any = 72.0%, Top-3 hit-any = 98.0%, and Top-5 hit-any = 99.0% across 100 test tracks with no prediction errors. Notebook 50 tested 60 external open-licence songs and produced Top-1 = 21.7%, Top-3 = 66.7%, and Top-5 = 83.3%. The difference between those results is not a problem to hide. It is the clearest guide for how the model should be presented.

| Decision area             | Management conclusion                                                                                                                                  |
|---------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|
| Project delivery          | The project produced notebooks, trained models, FMA test outputs, external-song outputs, documentation, and notebook PDFs.                             |
| Best use case             | Ranked genre recommendation, audio catalogue support, exploratory tagging, and retrieval assistance.                                                   |
| Formal FMA performance    | Notebook 49 reported Top-1 hit-any 72.0%, Top-3 98.0%, and Top-5 99.0% on the processed FMA test batch.                                                |
| External-song performance | Notebook 50 reported Top-1 21.7%, Top-3 66.7%, and Top-5 83.3% on 60 external open-licence songs.                                                      |
| Primary limitation        | The system is still affected by class imbalance, broad-label bias, and weaker rare-label performance.                                                  |
| Recommendation            | Present the system as a working academic prototype for ranked review. Keep production claims out until calibration and wider external testing improve. |

The recommended demonstration should therefore show Top-1, Top-3, and Top-5 predictions with scores and a low-confidence flag. That format fits the evidence and protects against overclaiming. It also shows the business value clearly: the model can reduce the number of genre choices a reviewer must inspect, while still leaving final judgement to a person or a catalogue rule.



Top-k review shows whether the accepted genre appears near the top, not only at Rank 1.

**Figure 1. Management-level summary of final Top-k evaluation results.**

Figure 1 gives the short version of the performance story. It should not be read as a claim that all genre tagging is solved. It shows that the FMA test outcome is strong under Top-k review, while the external test supports a more cautious deployment message.

## Business Problem and Opportunity

Large music libraries need genre tags for search, playlists, catalogue checks, and metadata review. Manual tagging is slow, and it depends heavily on the person doing the tagging. Automated genre prediction can reduce review time, but only if the output is treated with care. In this project, that means scores, Top-k lists, and low-confidence flags rather than a blind final label.

The need is addressed through a pipeline for both structured and unstructured music data. The system uses the FMA hierarchy and returns ranked suggestions for single tracks or external batches. This makes the output more useful for review and discovery than for blind automatic tagging.

## Project Delivery Overview

The delivery table separates completed capstone work from operational maturity. This distinction is important because a research prototype can be ready for submission and demonstration without being ready for unsupervised production use.

MongoDB was implemented as part of the project data engineering layer. It stored genre lookup records and multi-label track documents in the local fma\_capstone database. The final evaluation notebooks and Streamlit demo used exported CSV, JSON, NumPy, and saved model artefacts instead of live MongoDB reads because repeated audio inference and model execution were constrained by local CPU, memory, and disk I/O. In a fuller deployment, MongoDB would support catalogue lookup, prediction storage, and reviewer search.

| Delivery component        | Status    | Management value                                                                                                                             |
|---------------------------|-----------|----------------------------------------------------------------------------------------------------------------------------------------------|
| Data foundation           | Completed | FMA metadata, genre hierarchy, engineered features, raw audio paths, and external manifest data were organised for modelling and evaluation. |
| Model development         | Completed | The notebooks built and compared structured, audio CNN, and hybrid branches.                                                                 |
| Final FMA evaluation      | Completed | Notebook 49 provides the formal FMA test-split evidence.                                                                                     |
| External audio evaluation | Completed | Notebook 50 tests open-licence songs outside FMA.                                                                                            |
| Documentation             | Completed | Technical paper, graded proposal, readiness report, and file manifests are organised in the project folder.                                  |
| Software files            | Completed | All notebooks were exported to PDF with code and visible outputs.                                                                            |

## Decision Notes for Review

The main decision is how to present the model. The results support a ranked review tool, not an automatic final tagger. This is not a weakness in the submission. It is the operating boundary indicated by Notebook 49 and Notebook 50.

### Management Alternatives Considered

Three presentation options were considered. The first was to present the system as an automatic genre classifier. That option was rejected because the external Top-1 result did not support it. The second was to present only the FMA test result. That would have made the system look stronger, but it would have hidden the deployment risk shown by Notebook 50.

The third option, and the one recommended here, is to present the model as a ranked review tool. This option fits the evidence more closely. It allows the strong FMA Top-k results to be shown while still acknowledging that external audio needs review, calibration, and confidence controls.

Notebook 49 is the formal FMA benchmark. Notebook 50 is the more realistic external-song check. The gap between those two results is useful because it shows what changes when the model leaves the dataset it was trained and tested around.

## Key Findings

The key findings are connected. The audio branch became the strongest source of predictive signal because genre is heard through rhythm, timbre, instrumentation, and production style. Structured metadata and engineered features still had value, but they did not carry the same direct signal as the audio windows.

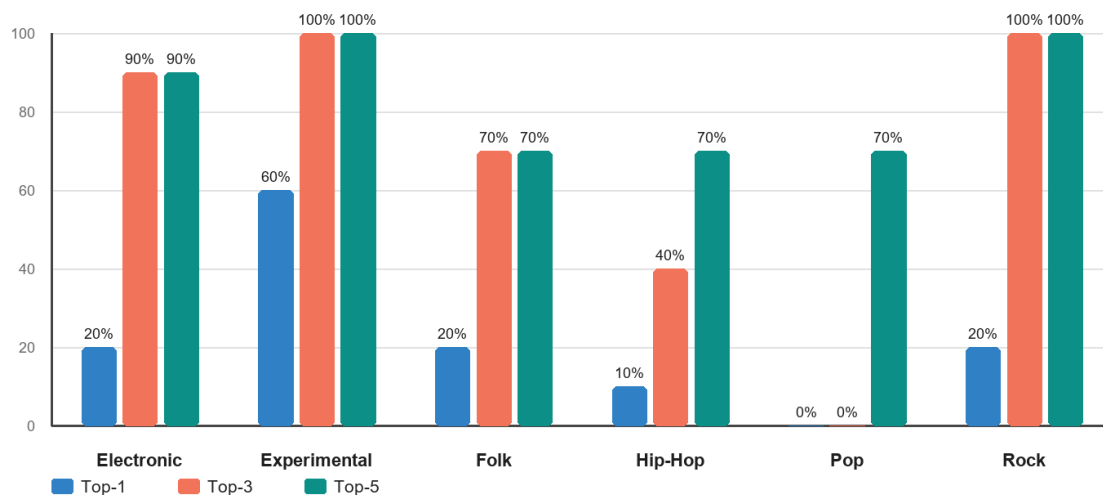
Hybrid fusion produced the best Candidate-150 benchmark because it combined the stronger audio signal with a smaller structured contribution. This supported the choice to keep three model branches in the report: structured modelling, audio CNN modelling, and hybrid fusion. The comparison also made the trade-off clearer. Audio gave stronger prediction power, while the structured branch gave an interpretable reference point.

Top-k evaluation gave the fairest management view. Music genre is often subjective and multi-label. A track may reasonably sit under several labels, especially when parent and child genres overlap. The final system is therefore more useful when it returns a ranked list rather than a single forced answer.

The corrected taxonomy plan uses 163 genre records. The model predicts 150 candidate labels directly, treats 10 rare labels through fallback logic, and keeps 3 labels as inventory-only records. This is an important management point because it prevents the project from claiming equal performance across labels that did not have equal training support.

External testing showed the main operational trade-off. Top-1 prediction was limited outside FMA, but Top-3 and Top-5 results remained useful. This means the system can support review and discovery, but it should not be used for automatic catalogue updates without more validation.

**External-Song Top-k Performance by Expected Genre**



Genre-level results show where external testing was stronger and where future data is needed.

**Figure 2. External-song Top-k performance by expected genre.**

Figure 2 is useful for management because it shows where future testing should focus. Uneven genre performance suggests that a larger and more varied external test set would be needed before operational rollout.

## What Changed During Final Testing

The main lesson from final testing was that evaluation setup matters as much as model choice. An empty test selection or mismatched audio window would have produced results that looked technical but were not meaningful. Correcting those issues improved the credibility of the final evidence.

The final stage was not just a clean run of an already finished model. Notebook 49 had to use corrected test selection so the evaluation used real records. Notebook 50 also had to handle full songs by using windows, because the trained audio model was built around 15-second inputs. These points are important because they show that the final evidence came from correction and review, not from simply accepting the first output.

These fixes make the final results more credible. They also explain why the report does not claim that the model predicts every song exactly. The stronger claim is narrower and better supported: it ranks useful genre candidates, especially when Top-3 and Top-5 results are considered.

## Operational Interpretation

For management purposes, this means the model should be shown with its uncertainty visible. A useful demo should not only display the top genre. It should also show the next strongest candidates and a low-confidence flag where the scores do not support a confident reading.

The model should be treated as a recommendation and review-support system. On FMA test records, results were strong under Top-3 and Top-5 review. On external songs, Top-5 stayed useful while Top-1 was modest. The practical reading is simple: likely genres can be surfaced for review, but final catalogue tags should not be assigned without review rules or a human check.

The practical value is strongest when the output is reviewed as a shortlist. A reviewer can inspect the first few labels, compare the confidence scores, and decide whether a broad label or a narrower subgenre is more appropriate. This is more realistic than asking the model to make a final judgement in every case.

| Use case                  | Readiness       | Comment                                                                                  |
|---------------------------|-----------------|------------------------------------------------------------------------------------------|
| Academic demonstration    | Ready           | The project is complete and backed by notebooks, outputs, documents, and software files. |
| Batch candidate tagging   | Partially ready | Useful for ranked genre suggestions, especially with Top-3 or Top-5 review.              |
| Automated final tagging   | Not yet ready   | Requires better calibration, confidence policy, and larger external validation.          |
| Rare-tail genre discovery | Limited         | Rare genres remain difficult due to data sparsity and should be treated cautiously.      |

## Risk and Limitation Summary

The limitation table is not included only as a compliance item. It summarises the main trade-offs that would affect real use: common genres perform better than rare ones, external songs may shift away from the training data, and broad labels can dominate the output.

The risk table should be read as a control list for the next stage of work. The main risks are not administrative; they come from the data and from the way genre labels behave in real music collections.

The risk table also explains why further work should focus on data and calibration before interface polish. A better interface would make the system easier to show, but it would not solve class imbalance or external domain shift. Those are modelling and evaluation issues first.

| Risk                             | Impact                                                       | Recommended control                                                                     |
|----------------------------------|--------------------------------------------------------------|-----------------------------------------------------------------------------------------|
| Class imbalance                  | Common genres receive stronger predictions than rare genres. | Use class balancing, augmentation, and hierarchy-aware evaluation in future work.       |
| External domain shift            | Songs outside FMA may not match the training distribution.   | Expand external test coverage and monitor confidence thresholds.                        |
| Broad-label bias                 | The model can over-select broad labels such as Experimental. | Use calibration, genre-family constraints, and post-processing review rules.            |
| Rare-tail weakness               | Sparse labels remain difficult to predict directly.          | Keep rare-tail output as supportive suggestions unless more labelled data is collected. |
| Single-label interpretation risk | A valid genre may appear below Rank 1.                       | Report Top-1, Top-3, and Top-5 results rather than one final label.                     |

## Management Recommendations

The first recommendation is to present the capstone as a completed applied data science prototype with ranked-prediction value. This wording is important because it matches the evidence. The notebooks show a functioning pipeline, saved outputs, and repeatable evaluation, but they also show that external Top-1 prediction is not strong enough for fully automatic tagging.

The second recommendation is to use Notebook 49 and Notebook 50 together during review. Notebook 49 should be used for the formal FMA benchmark because it tests the model against the project dataset. Notebook 50 should be used for the external audio demonstration because it shows how the same model behaves on new songs. Showing both creates a stronger and more honest presentation than showing only the best result.

The third recommendation is to avoid production-ready language. The safer and more accurate position is candidate ranking and review support. This does not weaken the project. It gives the model a practical role that is supported by the evidence and makes the limitations easier to manage.

The fourth recommendation is to build a small upload-based demo. The demo should return Top-1, Top-3, and Top-5 predictions, scores, and a low-confidence flag. This would make the value of the project easy to understand because reviewers could see how a song moves through the pipeline and why several candidates are shown.

The final recommendation is to treat calibration, rare-label learning, wider external testing, and a stronger serving interface as the next development priorities. These actions target the exact weaknesses found in final testing rather than adding unrelated features.

## Submission Readiness

The readiness table points to the main evidence files. A reviewer can use it to locate the final reports, notebook PDFs, saved models, final outputs, and external manifest without searching through the full project directory.

The readiness table records the files that should be available for review. It does not replace the technical paper, but it helps a reviewer locate the evidence behind the claims.

| Item                  | Location                                               | Status |
|-----------------------|--------------------------------------------------------|--------|
| Final technical paper | documentation/                                         | Ready  |
| Management report     | documentation/                                         | Ready  |
| Graded proposal       | documentation/Capstone_Project_Proposal_Graded.pdf     | Ready  |
| Notebook PDFs         | software artefacts/notebook_pdfs/                      | Ready  |
| Notebook PDF manifest | software artefacts/notebook_pdf_manifest.csv           | Ready  |
| Final model files     | models/                                                | Ready  |
| Final outputs         | outputs/notebook49_* and outputs/notebook50_*          | Ready  |
| External manifest     | data/external_audio_batch/external_audio_manifest.xlsx | Ready  |

## Recommended Demonstration Message

The demonstration should show one uploaded or selected song, then return Top-1, Top-3, and Top-5 predictions with scores. The presenter should explain that music genre can be multi-label and subjective. The strongest message is simple: the model helps narrow the review choices rather than replacing judgement.

## Conclusion

The capstone package is complete for academic submission and demonstration. Its strongest value is a ranked prediction workflow that uses audio and supporting metadata to narrow likely genre choices. The final evidence supports this use case, especially when Top-3 and Top-5 outputs are considered.

The management decision should therefore be to present the system with clear operating boundaries. It should be shown as a review-support model, with confidence scores and low-confidence flags visible. This protects the work from overclaiming while still showing the technical achievement.

The next practical step is a small demo interface that accepts an audio file and returns ranked genre suggestions. That demo would connect the technical work to a reviewer-facing workflow and make the capstone easier to assess during presentation.